Continuous Robot Path Planning for Human-inspired Information Gathering

Michael Leong, Shipeng Liu, Guangyao Shi, Prof. Feifei Qian
USC Viterbi Department of Electrical & Computer Engineering
Robot Locomotion and Navigation Dynamics Lab

USC Viterbi School of Engineering

RoboLAND
Robot Locomotion and Navigation Dynamics

Introduction

- Geoscientists require data collection from various environments to support their hypothesis (i.e. terrain strength, moisture levels).
- Robots are able to explore unknown environments, but can follow routes or produce results undesired by real human scientists.

Research Question

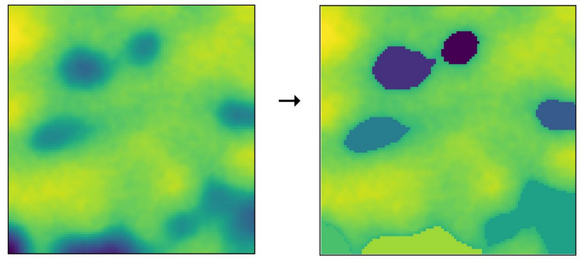
How can we make a system to utilize quadruped robots to collect data and receive feedback from humans?

Objectives

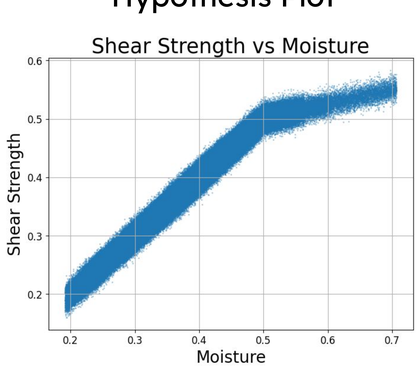
- Utilize two human-inspired path-planning methods: **Micro Gradient** and **Zone Coverage.**
- Implement algorithms to **LASSIE Decision Making** – human study website – to evaluate the effectiveness of the two human inspired algorithms. Humans provide feedback while robot performs them in real-time.
- Want to see if the algorithms are preferred by the human and can actually help the scientist evaluate their hypothesis more quickly.

Methods

Shear Strength Map → Regions Filtered

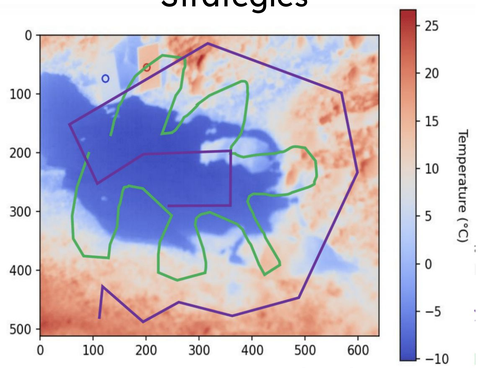


Hypothesis Plot



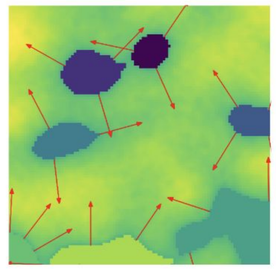
- Plotted points are sampled data
- Refer to plot to evaluate whether hypothesis is true

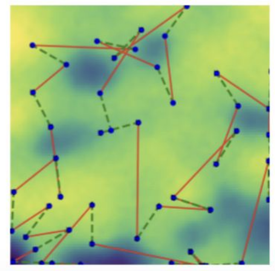
Example of Human Inspired Strategies



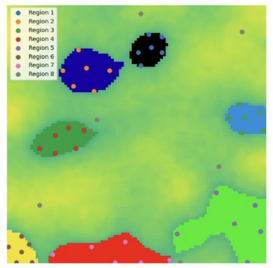
Zone Coverage Methods
Micro Gradient Methods
Information-gathering on spatial wise

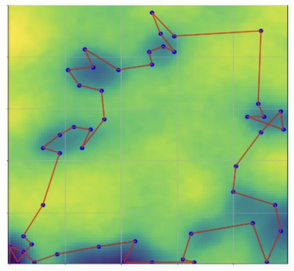
Gradient Directions on Edges (Micro Gradient) → Plan path across gradients



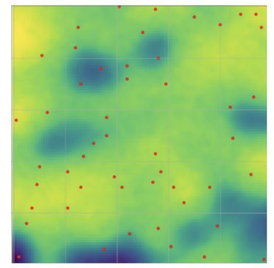


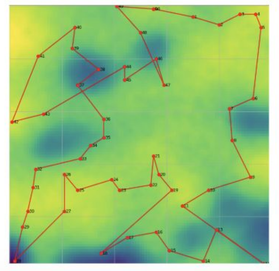
Randomly sampled points within regions (Zone Coverage) → Plan path across all regional points





Maximized the information coverage within the limited budget (Baseline) → Plan path across all points





Next Steps

- Implement these path-planning algorithms to **LASSIE Decision Making** and have geoscientists evaluate the effectiveness of these algorithms.
- Test in field trip in various outdoor environments (i.e. desert, beach, park, etc).

Acknowledgements

This work is supported by the NASA Planetary Science and Technology Through Analog Research (PSTAR) program, Award Number: 80NSSC22K1313. I'd like to thanks my mentors Shipeng Liu, Guangyao Shi, and Prof. Feifei Qian for their valuable advisement.